



DIGITAL COMMUNICATION LABORATORY

EXPERIMENT 06

Digital Line Coding & Power Spectral Density

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Software: Python / Google Colab

Python • Manual encoding • Welch PSD • DC • Bandwidth • Polarity • Self-clocking

Common 8-bit validation sequence: 10110010

Aim and required tasks

Generate six codes

Use the same bit sequence for all six line-code waveforms, without toolbox encoders.

Estimate PSD

Use Welch's method and normalize each PSD for comparison.

Compare

DC component, bandwidth, polarity and self-clocking capability.

Codes: Unipolar NRZ, Polar NRZ-L, Polar RZ, Manchester, Differential Manchester and AMI.

Validate the exact 8-bit word 10110010 bit-by-bit.

Measure average level and running digital sum.

Test a long run of identical bits to expose synchronization limitations.

Definitions and conventions

Unipolar NRZ

$1 = +A$; $0 = 0$. Non-zero DC for data with a non-zero fraction of 1s.

Polar NRZ-L

$1 = +A$; $0 = -A$. Two-level polar signaling; transition density depends on data.

Polar RZ

Each bit starts at $\pm A$ and returns to 0 halfway through the bit.

Manchester

$1 = -A \rightarrow +A$; $0 = +A \rightarrow -A$. Always one mid-bit transition.

Differential Manchester

$0 =$ transition at bit start; $1 =$ no start transition. Always a mid-bit transition.

AMI

$0 = 0$; successive 1s alternate $+A$ and $-A$. Bipolar and DC-suppressing.

Convention note

Manchester polarity conventions vary. Differential Manchester also has convention variants; the definitions above are exactly the conventions used in the Python code.

Python — manual encoder implementation

```
bits8 = np.array([1,0,1,1,0,0,1,0])
A = 1.0
Ns = 100
Rb = 1.0
Fs = Rb * Ns

def unipolar_nrz(bits, Ns, A=1.0):
    return np.repeat(bits.astype(float)*A, Ns)

def polar_nrz_l(bits, Ns, A=1.0):
    return np.repeat(np.where(bits==1,A,-A), Ns)

def polar_rz(bits, Ns, A=1.0):
    x = np.zeros(len(bits)*Ns); h = Ns//2
    for k,b in enumerate(bits):
        x[k*Ns:k*Ns+h] = A if b==1 else -A
    return x

def manchester(bits, Ns, A=1.0):
    x = np.zeros(len(bits)*Ns); h = Ns//2
    for k,b in enumerate(bits):
        if b==1:
            x[k*Ns:k*Ns+h] = -A
            x[k*Ns+h:(k+1)*Ns] = A
        else:
            x[k*Ns:k*Ns+h] = A
            x[k*Ns+h:(k+1)*Ns] = -A
    return x
```

Python — Differential Manchester and AMI

```
def differential_manchester(bits, Ns, A=1.0):
    x = np.zeros(len(bits)*Ns); h = Ns//2
    level = A
    for k,b in enumerate(bits):
        if b==0:                # start transition
            level = -level
            x[k*Ns:k*Ns+h] = level
            level = -level      # mandatory mid-bit transition
            x[k*Ns+h:(k+1)*Ns] = level
    return x

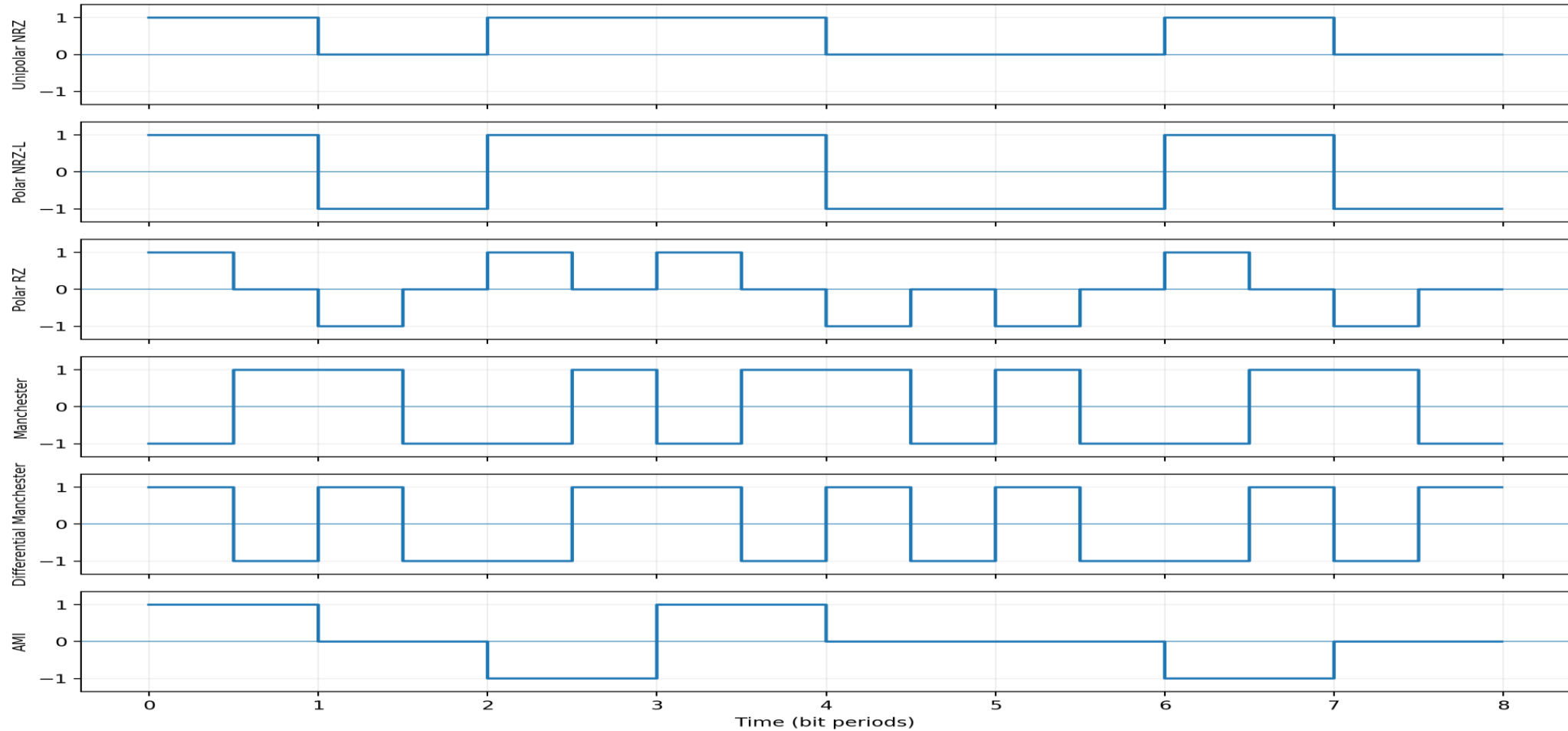
def ami(bits, Ns, A=1.0):
    x = np.zeros(len(bits)*Ns); last = -A
    for k,b in enumerate(bits):
        if b==1:
            last = -last
            x[k*Ns:(k+1)*Ns] = last
    return x

signals = {
    "Unipolar NRZ": unipolar_nrz(bits8,Ns,A),
    "Polar NRZ-L": polar_nrz_l(bits8,Ns,A),
    "Polar RZ": polar_rz(bits8,Ns,A),
    "Manchester": manchester(bits8,Ns,A),
    "Differential Manchester": differential_manchester(bits8,Ns,A),
    "AMI": ami(bits8,Ns,A)
}
```

Output 1 — six line-code waveforms

All waveforms use exactly the same 8-bit sequence: 10110010

Six manually generated line codes — 10110010



Welch PSD — correct experimental procedure

```
from scipy.signal import welch

# Same 8-bit pattern, repeated only to create a longer record
# for a stable Welch estimate:
psd_bits = np.tile(bits8, 128)

f, Pxx = welch(
    x, fs=Fs, window="hann",
    nperseg=1024, noverlap=512,
    nfft=4096, detrend=False,
    return_onesided=False,
    scaling="density"
)

Pnorm = Pxx / np.max(Pxx)
Pxx_dB = 10*np.log10(Pnorm + 1e-12)
```

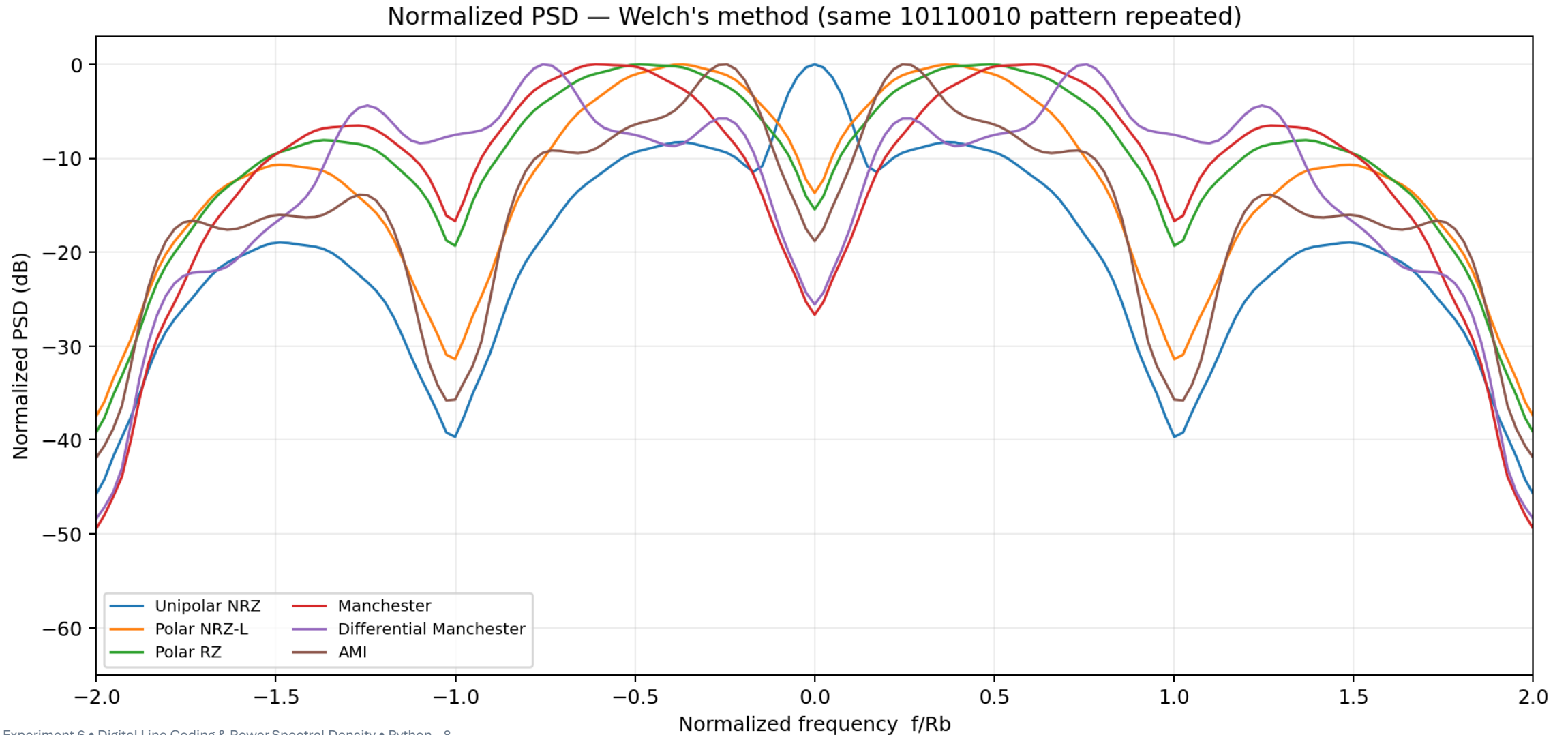
Same data pattern

The PSD record is 10110010 repeated 128 times. Therefore every code is driven by the same data pattern; no different random data is used.

DC must be retained

`detrend=False` is intentional because removing the mean would remove the DC component needed for comparison.

Output 2 — normalized PSD using Welch's method



DC component — average signal level

Exact 8-bit sequence: 10110010

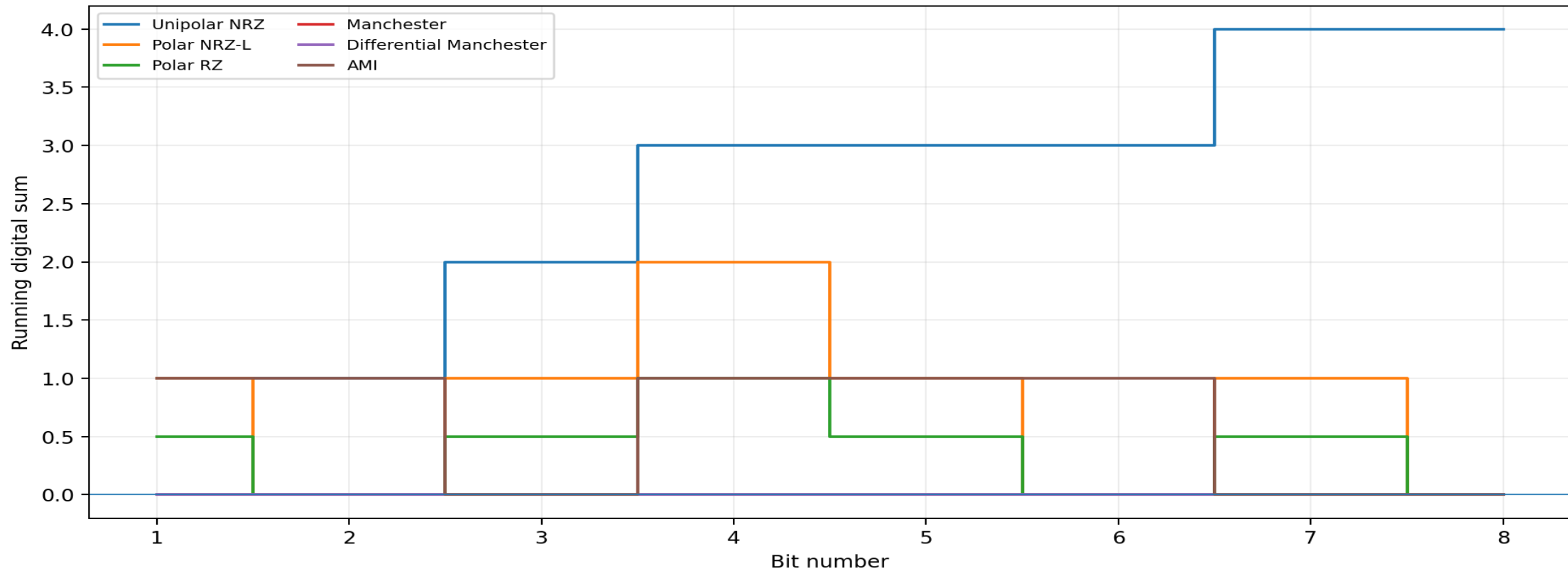
Line code	Average level	Correct interpretation
Unipolar NRZ	+0.5000	Non-zero DC: 4 ones out of 8 → +0.5
Polar NRZ-L	+0.0000	Zero for this balanced word
Polar RZ	+0.0000	Zero for this word
Manchester	+0.0000	Zero; each bit contains equal +A/−A halves
Differential Manchester	+0.0000	Zero; equal positive/negative half-bit durations
AMI	+0.0000	Zero; four 1s alternate polarity

Important

Average level is the DC indicator for this finite data record. General DC behavior is data-dependent.

Output 3 — Running Digital Sum

Running Digital Sum — 10110010



Interpretation

The RDS is the cumulative sum of per-bit mean levels. Bounded/zero-ending RDS indicates good DC balance for this word.

Mandatory 8-bit manual validation

Per-bit mean levels for 10110010; verify the actual mid-bit/boundary transitions from the waveform

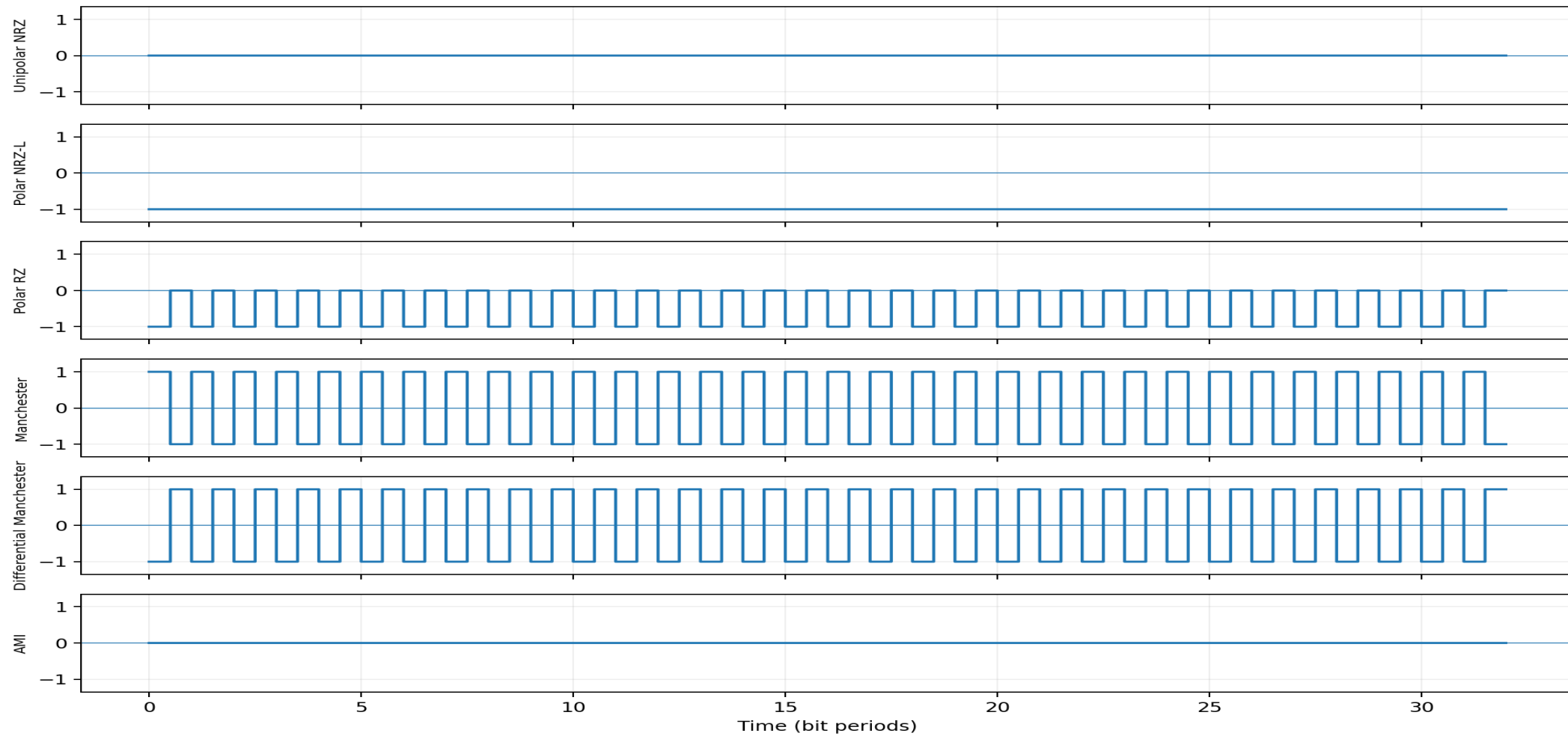
Code	b1	b2	b3	b4	b5	b6	b7	b8	Final RDS
Unipolar NRZ	+1.0	+0.0	+1.0	+1.0	+0.0	+0.0	+1.0	+0.0	+4.0
Polar NRZ-L	+1.0	-1.0	+1.0	+1.0	-1.0	-1.0	+1.0	-1.0	+0.0
Polar RZ	+0.5	-0.5	+0.5	+0.5	-0.5	-0.5	+0.5	-0.5	+0.0
Manchester	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
Differential Manchester	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
AMI	+1.0	+0.0	-1.0	+1.0	+0.0	+0.0	-1.0	+0.0	+0.0

- Unipolar NRZ: 1 → +1; 0 → 0.
- Polar NRZ-L: 1 → +1; 0 → −1.
- Polar RZ: ±1 for first half of the bit, then 0.
- Manchester: mandatory middle transition in every bit.
- Differential Manchester: mandatory middle transition plus data-dependent start transition.
- AMI: 1s alternate +1/−1; 0s remain at 0.

Long-run identical-bit test

32 consecutive zeros

Long-run test — 32 identical zeros



Engineering comparison — correct theoretical interpretation

Code	DC component	Relative bandwidth	Polarity	Self-clocking
Unipolar NRZ	High / non-zero	Low	Unipolar	Poor
Polar NRZ-L	Data-dependent	Low	Polar ($\pm A$)	Poor
Polar RZ	Data-dependent	Higher than NRZ	Polar ($\pm A$, return to 0)	Data-dependent; timing improved
Manchester	Ideally zero	High	Polar ($\pm A$)	Excellent
Differential Manchester	Ideally zero	High	Two-level differential	Excellent
AMI	Ideally zero for balanced 1s	Moderate	Bipolar ($+A/-A/0$)	Limited / data-dependent

Bandwidth note

These are relative bandwidth classifications. Exact numerical bandwidth depends on the chosen definition (first spectral null, occupied bandwidth, etc.) and PSD threshold.

Complete Python workflow

```
# 1. Define common data
bits8 = np.array([1,0,1,1,0,0,1,0])

# 2. Generate all six codes manually
signals = {...}

# 3. Plot exact 8-bit validation waveforms
t = np.arange(len(bits8)*Ns)/Fs
plt.step(t, x, where="post")

# 4. Use the same data pattern repeatedly for Welch
psd_bits = np.tile(bits8, 128)

# 5. Welch PSD
f, Pxx = welch(x, fs=Fs, window="hann",
               nperseg=1024, noverlap=512,
               nfft=4096, detrend=False,
               return_onesided=False)

# 6. Normalize
Pnorm = Pxx / np.max(Pxx)
Pxx_dB = 10*np.log10(Pnorm + 1e-12)

# 7. DC / RDS
average_level = np.mean(x)
RDS = np.cumsum(bit_means)

# 8. Long-run test
long_bits = np.zeros(32,dtype=int)
```

Observation, interpretation and conclusion

The six codes are generated manually from the same 8-bit pattern, so the waveform comparison is a direct comparison of coding rules.

Unipolar NRZ has a non-zero DC component for 10110010 because four of eight bits are 1.

Polar NRZ-L, Polar RZ, Manchester, Differential Manchester and AMI have zero average for this particular balanced word.

Manchester and Differential Manchester have mandatory mid-bit transitions, giving excellent self-clocking but higher bandwidth.

Polar RZ provides timing transitions for 1 bits but has no transition during a run of zeros; its synchronization is therefore data-dependent.

AMI alternates the polarity of successive 1s and suppresses DC, but a long run of zeros contains no pulses and can lose timing information.

Welch's PSD confirms the expected spectral trade-off: NRZ is relatively narrowband, while RZ/Manchester-family signals have greater high-frequency content.

All measured observations are consistent with standard line-coding theory.

Final conclusion

Line coding is a trade-off among DC balance, bandwidth, polarity behavior and clock recovery.